

Technology Stack

Date	04 July 2026
Team ID	XXXXXX
Project Name	Smart-Lender-Loan-Eligibility-Prediction
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, front-end tools, back-end tools, development tool and version control that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	HTML5, CSS3, JavaScript, Bootstrap 5	Responsive and user-friendly interface to enter loan applicant details and display prediction results.

2	Backend / Server-Side	Python, Flask	Flask handles HTTP requests, processes user input, loads the trained ML model, and returns loan eligibility predictions efficiently.
3	Database / Data Storage	CSV Dataset, Pickle (.pkl) Model	CSV is used for training data, while the trained ML model is stored in .pkl file for fast predictions
4	Machine Learning	Scikit-learn, Random Forest..	Used to train, test, evaluate and predict loan eligibility using machine learning algorithms
5	Data Preprocessing	Pandas, NumPy	Used for data cleaning, preprocessing, feature handling and numerical computations

6	Development Tool / Version Control	Jupyter Notebook, Visual Studio Code, Git, Github	JN is used for model development; VS Code is used for Flask application development; Used for version control, project management and upload code in github
---	---	---	--